

EXERCISE 5.6 - SATURATION OSCILLATOR

5.6 In this exercise, adapted from [Phillips and Harbor, 1988], let us consider the system of Figure 5.32, which is typical of the dynamics of electronic oscillators used in laboratories, with

$$G(p) = \frac{-5p}{p^2 + p + 25}$$

Use describing function analysis to predict whether the system exhibits a limit cycle, depending on the value of the saturation level k . In such cases, determine the limit cycle's frequency and amplitude.

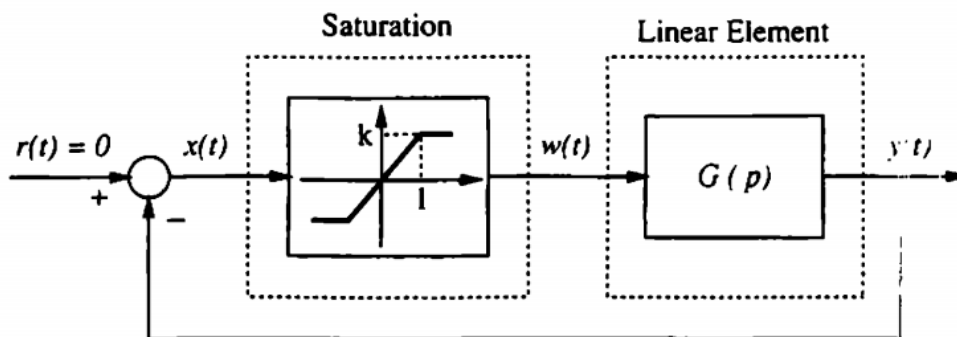


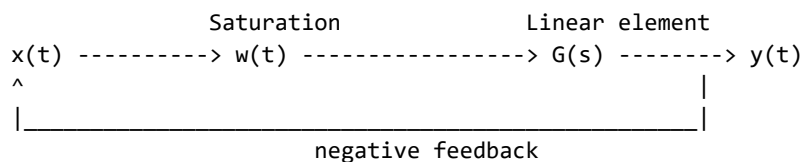
Figure 5.32 : Dynamics of an electronic oscillator

Interpret intuitively, by assuming that the system is started at some small initial state, and noticing that $y(t)$ can stay neither at small values (because of instability) nor at saturation values (by applying the final value theorem of linear control).

Solution:

Describing Function Analysis

System:



$$r(t) = 0$$

Linear element:

$$G(s) = \frac{-5s}{s^2 + s + 25}$$

$$s^2 + s + 25$$

Saturation characteristic:

$$w = k \cdot x \text{ for } x \leq 1 \quad w = k \cdot \text{sign}(x) \text{ for } x > 1$$

Therefore, k is the saturation output level and the breakpoint is $x = 1$.

This script determines: 1. whether a limit cycle exists, 2. its frequency, 3. its amplitude, 4. its stability, using the describing function technique.

```
clear;
clc;
close all;
```

1. Saturation Level

Change k to study different cases.

Examples:

$k < 0.2$: no limit cycle

$k = 0.2$: marginal case

$k > 0.2$: stable limit cycle

```
k = 1;
```

2. Linear Element

```
num = [-5 0];
den = [1 1 25];

Gs = tf(num,den);
```

3. Describing Function of the Saturation

For a sinusoidal input:

$$x(t) = A \cdot \sin(w \cdot t)$$

If $A \leq 1$, the saturation does not become active:

$$N(A) = k$$

If $A > 1$:

$$N(A) = \frac{2k}{\pi} \left[\arcsin\left(\frac{1}{A}\right) + \sqrt{1 - \frac{1}{A^2}} \right]$$

$$\frac{pi}{A} = \frac{A^2}{A^2}$$

The saturation is memoryless, so $N(A)$ is real and does not depend on w .

4. Limit-Cycle Condition

The describing-function condition is:

$$G(jw) * N(A) = -1$$

Since $N(A)$ is real and positive, $G(jw)$ must be negative real.

For

$$-5jw$$

$$G(jw) = \frac{-5jw}{25 - w^2 + jw}$$

$$25 - w^2 + jw$$

$$\text{imag}(G(jw)) = 0 \text{ gives:}$$

$$w = 5 \text{ rad/s}$$

At $w = 5 \text{ rad/s}$:

$$G(j5) = -5$$

Therefore:

$$(-5) * N(A) = -1$$

$$N(A) = 0.2$$

```
w_limit = 5;
N_required = 1/5;
```

5. Decide Whether a Limit Cycle Exists

```
fprintf('\nEXERCISE 5.6 - DESCRIBING FUNCTION ANALYSIS\n');
```

```
EXERCISE 5.6 - DESCRIBING FUNCTION ANALYSIS
```

```
fprintf('-----\n');
```

```
-----
```

```
fprintf('Saturation level k = %.4f\n',k);
```

```
Saturation level k = 1.0000
```

```
fprintf('Required describing-function value = %.4f\n\n',N_required);
```

```
Required describing-function value = 0.2000
```

```

if k < N_required

    fprintf('k < 0.2\n');
    fprintf('Since N(A) <= k < 0.2 for every amplitude,\n');
    fprintf('G(jw)N(A) cannot reach -1.\n\n');
    fprintf('RESULT: NO LIMIT CYCLE is predicted.\n');
    fprintf('The small-signal closed-loop system is stable.\n\n');

    limit_cycle_exists = false;
    marginal_case = false;
    A_limit = NaN;

elseif abs(k-N_required) < 1e-10

    fprintf('k = 0.2\n');
    fprintf('For A <= 1, N(A) = 0.2 exactly.\n');
    fprintf('The small-signal closed-loop poles are on the imaginary axis.\n\n');
    fprintf('RESULT: This is a MARGINAL CASE.\n');
    fprintf('There is no unique isolated limit cycle predicted by the describing
function.\n\n');

    limit_cycle_exists = false;
    marginal_case = true;
    A_limit = NaN;

else

    fprintf('k > 0.2\n');
    fprintf('A limit cycle is predicted.\n\n');

    limit_cycle_exists = true;
    marginal_case = false;

```

```

k > 0.2
A limit cycle is predicted.

```

6. Find the Limit-Cycle Amplitude

For $A > 1$:

$$N(A) = 0.2$$

Solve this equation numerically.

```

Nfun = @(A) (2*k/pi)*(asin(1./A) + (1./A).*sqrt(1 - 1./A.^2));

equation = @(A) Nfun(A) - N_required;

```

```

% Find an upper bound where the function changes sign.
A_high = 2;

while equation(A_high) > 0
    A_high = 2*A_high;
end

% Solve for the amplitude.
A_limit = fzero(equation,[1+1e-6 A_high]);

```

7. Display Limit-Cycle Result

```

fprintf('LIMIT CYCLE PREDICTION\n');
fprintf('Amplitude A = %.4f\n',A_limit);
fprintf('Angular frequency w = %.4f rad/s\n',w_limit);
fprintf('Frequency f = %.4f Hz\n',w_limit/(2*pi));
fprintf('Period T = %.4f s\n\n',2*pi/w_limit);

```

```
end
```

```

LIMIT CYCLE PREDICTION
Amplitude A = 6.3397
Angular frequency w = 5.0000 rad/s
Frequency f = 0.7958 Hz
Period T = 1.2566 s

```

8. Nyquist Plot: $G(j\omega)$ and $-1/N(A)$

```

w = linspace(0.01,20,4000);

s = 1i*w;

Gjw = (-5*s)./(s.^2 + s + 25);

% Amplitude vector for the describing-function curve.
if limit_cycle_exists
    Amax = max(10,1.5*A_limit);
else
    Amax = 10;
end

A = linspace(0.05,Amax,1500);

N = zeros(size(A));

for i = 1:length(A)

    if A(i) <= 1

```

```

        N(i) = k;

    else

        N(i) = (2*k/pi)*(asin(1/A(i)) + (1/A(i))*sqrt(1 - 1/A(i)^2));

    end

end

minus_inv_N = -1./N;

figure('Name','Exercise 5.6 - Describing Function', 'NumberTitle','off');

% Positive-frequency Nyquist branch.
plot(real(Gjw),imag(Gjw),'LineWidth',2);
hold on;

% Negative-frequency branch.
plot(real(Gjw),-imag(Gjw),'LineWidth',1);

% -1/N(A) lies on the negative real axis.
plot(minus_inv_N,zeros(size(minus_inv_N)),'--','LineWidth',2);

% Important point of the linear element.
plot(-5,0,'ko','MarkerFaceColor','k');

% Important point of the linear element.
plot(-5,0,'ko','MarkerFaceColor','k');

if limit_cycle_exists

    % Limit cycle point
    plot(-5,0,'ro','MarkerFaceColor','r');

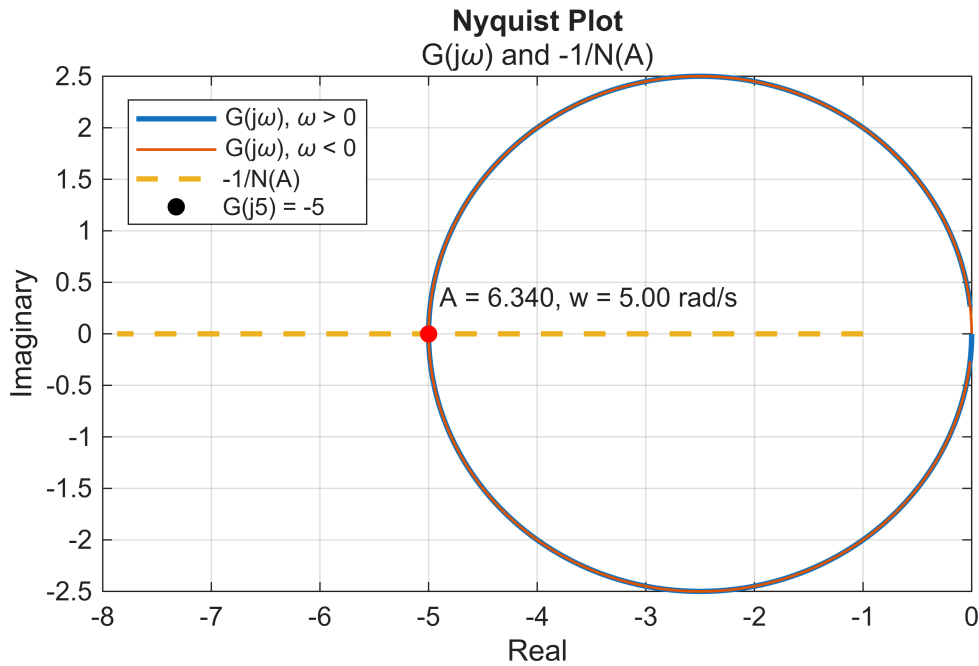
    % Show amplitude and frequency near the intersection point
    text(-5 + 0.1, 0.35, sprintf('A = %.3f, w = %.2f rad/s',A_limit,w_limit));

end

grid on;
xlabel('Real');
ylabel('Imaginary');
title('Nyquist Plot');
subtitle('G(j\omega) and -1/N(A)');

legend('G(j\omega), \omega > 0', 'G(j\omega), \omega < 0', '-1/N(A)', 'G(j5) = -5',
'Location','best');

```



9. Plot $G(j\omega)N(A)$ for Different Amplitudes

If limit cycle exists:

Choose amplitudes below, at, and above the predicted limit cycle.

else:

Example amplitudes when there is no unique limit cycle.

```
figure('Name','Exercise 5.6 - Limit Cycle Detection', 'NumberTitle','off');

hold on;

if limit_cycle_exists

    A_values = [1, 0.75*A_limit, A_limit, 1.25*A_limit];

else

    A_values = [0.5 1 2 5];

end

legend_text = cell(1,length(A_values)+1);

for m = 1:length(A_values)

    A0 = A_values(m);
```

```

if A0 <= 1

    N0 = k;

else

    N0 = (2*k/pi)*(asin(1/A0) + (1/A0)*sqrt(1 - 1/A0^2));

end

L = Gjw*N0;

plot(real(L),imag(L),'LineWidth',1.5);

legend_text{m} = sprintf('A = %.3f', A0);

end

```

Critical point

```

plot(-1,0,'ko','MarkerFaceColor','k');
legend_text{end} = '(-1,0)';

```

Show limit cycle frequency near the critical point

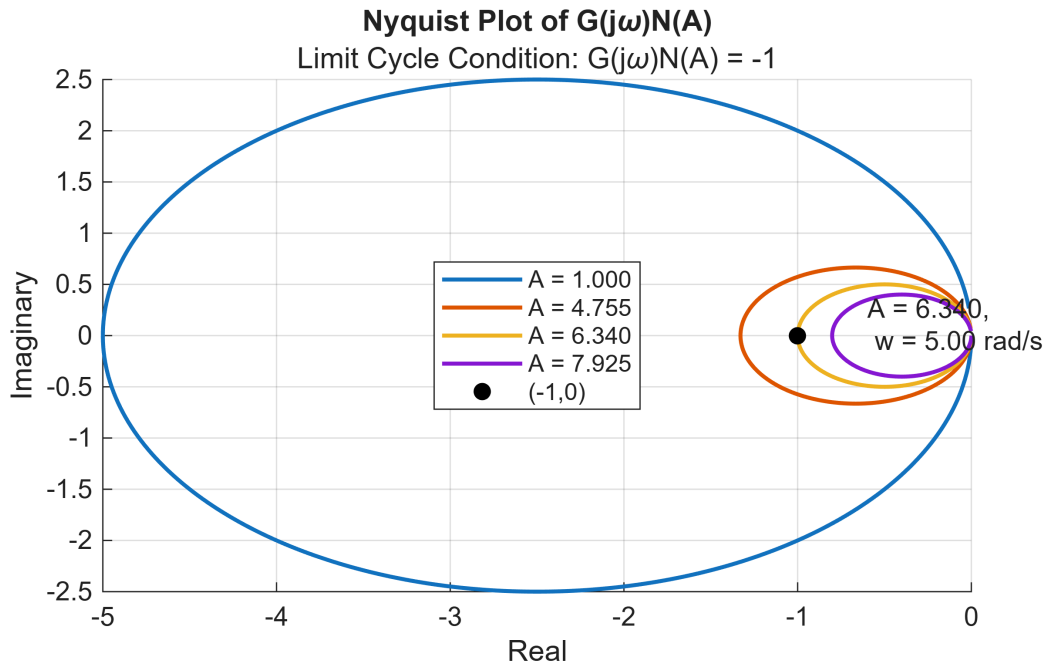
```

if limit_cycle_exists
    text(-1 + 0.4, 0.1, sprintf('A = %.3f,\n w = %.2f rad/s',A_limit,w_limit));
end

grid on;
xlabel('Real');
ylabel('Imaginary');
title('Nyquist Plot of G(j\omega)N(A)');
subtitle('Limit Cycle Condition: G(j\omega)N(A) = -1');

legend(legend_text,'Location','best');

```

10. Stability of the Predicted Limit Cycle

If Limit Cycle exists:

The equivalent closed-loop characteristic equation is:

$$s^2 + (1 - 5N(A))s + 25 = 0$$

Therefore:

effective damping coefficient = $1 - 5N(A)$

Below the limit-cycle amplitude:

$N(A) > 0.2 \rightarrow \text{damping} < 0 \rightarrow \text{amplitude grows.}$

Above the limit-cycle amplitude:

$N(A) < 0.2 \rightarrow \text{damping} > 0 \rightarrow \text{amplitude decreases.}$

Hence the limit cycle is stable.

```
if limit_cycle_exists

    dA = 0.1*A_limit;

    A_low = max(1+1e-6,A_limit-dA);
    A_high = A_limit+dA;

    N_low = Nfun(A_low);
    N_high = Nfun(A_high);
```

```

damping_low = 1 - 5*N_low;
damping_high = 1 - 5*N_high;

fprintf('LIMIT CYCLE STABILITY\n');
fprintf('-----\n');

fprintf('Below the limit cycle:\n');
fprintf('A = %.4f\n',A_low);
fprintf('N(A) = %.4f\n',N_low);
fprintf('1 - 5N(A) = %.4f\n',damping_low);

if damping_low < 0
    fprintf('Amplitude increases toward the limit cycle.\n\n');
else
    fprintf('Amplitude does not increase toward the limit cycle.\n\n');
end

fprintf('Above the limit cycle:\n');
fprintf('A = %.4f\n',A_high);
fprintf('N(A) = %.4f\n',N_high);
fprintf('1 - 5N(A) = %.4f\n',damping_high);

if damping_high > 0
    fprintf('Amplitude decreases toward the limit cycle.\n\n');
else
    fprintf('Amplitude does not decrease toward the limit cycle.\n\n');
end

if damping_low < 0 && damping_high > 0
    fprintf('Therefore, the predicted limit cycle is STABLE.\n\n');
else
    fprintf('The stability test is inconclusive.\n\n');
end

end

```

```

LIMIT CYCLE STABILITY
-----
Below the limit cycle:
A = 5.7057
N(A) = 0.2220
1 - 5N(A) = -0.1100
Amplitude increases toward the limit cycle.
Above the limit cycle:
A = 6.9737
N(A) = 0.1820
1 - 5N(A) = 0.0902
Amplitude decreases toward the limit cycle.
Therefore, the predicted limit cycle is STABLE.

```

11. Final Analytical Result

```
fprintf('FINAL RESULT\n');
```

FINAL RESULT

```
fprintf('-----\n');
```

```
fprintf('k < 0.2 : no limit cycle is predicted.\n');
```

k < 0.2 : no limit cycle is predicted.

```
fprintf('k = 0.2 : marginal case; no unique isolated limit cycle.\n');
```

k = 0.2 : marginal case; no unique isolated limit cycle.

```
fprintf('k > 0.2 : stable limit cycle is predicted.\n');
```

k > 0.2 : stable limit cycle is predicted.

```
fprintf('          w = 5 rad/s.\n');
```

w = 5 rad/s.

```
fprintf('          A is found from N(A) = 0.2.\n');
```

A is found from N(A) = 0.2.

```
if limit_cycle_exists
    fprintf('For k = %.4f, A = %.4f.\n',k,A_limit);
end
```

For k = 1.0000, A = 6.3397.